BODA KALYAN SINGH

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EDUCATION

YEAR	DEGREE	INSTITUTE	CGPA/MARKS
2025	M. TECH/Earth system science&tech	INDIAN INSTITUTE OFTECHNOLOGY KHARAGPUR	8.84
2022	B. TECH/Civil	JNTUH COLLEGE OF ENGINEERING MANTHANI	6.44
2018	TSBIE	SRI CHAITANYA JUNIOR COLLEGE	97.4
2016	TSBSE	GOOD SHEPHERD SCHOOL	9.7

EXPERIENCES

Full Stack AI Developer | JUTEQ ***[June 2025-Ongoing]***

At JUTEQ, I contributed to the development of automotive agents for intelligent auto-reply and email response and SMS responses systems using Vertex AI and large language models. I worked on building ReAct-style agents and Agentic RAG pipelines, integrating LLM orchestration frameworks like LangChain and CrewAI, and deploying scalable AI services with Node.js, NestJS, and React. My focus included system observability, AI safety, and real-world deployment optimization.

Founder & Chief Developer | CampusBuzz APP | Link ***[Aug - 2024]***

Launched a dynamic web and mobile platform for the IIT community, featuring campus Q&A, alumni forums, mentorship programs, student marketplace, JEE predictor, and an AI-powered chatbot. Achieved 550+ organic downloads and 250+ signups without marketing. Led and mentored a team of 30+ interns across development, design. Recognized among the Top 100 startups by Inflection Point Ventures and acknowledged by the Entrepreneurship Cell, IIT Kharagpur. Available on Google Play

Co – Founder | Olive Orange | AI-Powered EdTech Innovation | Link ***[Dec – 2024]***

Designed and developed an AI-powered education platform leveraging Large Language Models (LLMs) to transform teaching and learning experiences. The platform equips educators with smart tools for lesson planning, test automation, and rich content resources, while offering students adaptive assessments, personalized coaching, and multilingual support. It also features curriculum alignment, robust privacy settings, and an intuitive parent dashboard to provide real-time insights into student progress

SKILLS

Programming Languages	*Python, SQL, HTML,CSS, JavaScript, MATLAB*
Frameworks & Tools	*TensorFlow, PyTorch, LangChain, LlamaIndex, Streamlit, Flask, Django, FastAPI, Git, VS Code, Github, Ollama, Pandas,Numpy,Oops Chainlit, LangSmith and GPT4ALL. React, n8n,Hugging Face Transformers, LangFlow, Flowise, Haystack, Postman*
Generative AI Technologies	*Llama3, Gemma, GPT-4,Anthropic Claude (3, 3 Opus, Sonnet), Gemini, ElevenLabs, Smallest.ai, Luma AI, Leonardo.Ai, Ideogram, Proplexity, Tavily, Ollama, GPT4ALL*
Vector Databases	*ChromaDB, Pinecone, FAISS, Redis, Google's Matching Engine (Vector Search)*
Database Management	*Firestore, Cloud SQL, Memorystore, Cloud Bigtable, Redis, PostgreSQL*
Deployment Platforms	*GCP,Azure, Cloud Functions, APP Engine, AWS Bedrock, AWS (EC2, Lambda)*
AI/ML Techniques	*Fine-tuning with custom data, Vector Embedding, Natural Language Processing, Machine Learning, Deep learning, Neural Networks, MLOPS, DevOps, Dockers, Kubernetes, CI/CD Pipeline, Large Language Models, Advanced RAG Pipeline.*
AI Agents & Automation	*Agent building (AutoGPT, BabyAGI) Multi-agent systems MCP (Model Context Protocol) Agentic frameworks (LlamaIndex agents, CrewAI, AutoGen) Autonomous agents Tool integration (Tavily, Perplexity) LangGraph*
API Development	*RESTAPI design and implementation,Postman,Swagger/OpenAPI ,API authentication, API gateways, GraphQL, API versioning, Webhook management*
Prompt Engineering	*Instruction-based Prompting, Contextual Prompting, Few-shot Learning, Chain-of-Thought Prompting, Role-based Prompting, Zero-shot Prompting,System Prompting Persona Design Guard Rails ReAct Prompting Self-consistency Techniques*
LLM Evaluation Metrics	*Relevance Metrics: Perplexity, BLEU, ROUGE. Alignment Metrics: Truthfulness, Safety, Fairness. Task-Specific Metrics: GLUE, SQuAD, Ragas TruLens,DeepEval*
Data Visualization	*Power BI, Matplotlib, Seaborn, Excel, Tableau, Plotly*
Other Known software	*Android studio, UI/UX, Figma, QGIS,CAD,SNAP,GEE, PWA ,Jira, Confluence*
Soft Skills	*Analytical thinking, Problem-solving, Teamwork, Leadership, Time management*

INTERNSHIPS

➢ **Full-Stack GenAI Trainee at NagentAI** ***Feb 2025 – May 2025***

- Developed and deployed 50+ intelligent AI agents as a Full-Stack GenAI Engineer at NagentAI, leveraging diverse LLM technologies (OpenAI GPT-4/3.5, Claude 3 models, Llama3, Gemini, Mistral, Gemma, Flux, Smallest AI) and agent frameworks. Created diverse solutions including ad video generation systems, SEO content optimizers, automated company research tools,

LinkedIn post generators with scheduling capabilities, dueling capabilities, and many more. Implemented advanced RAG pipelines, vector databases (ChromaDB, FAISS), and multimodal AI integration (ElevenLabs, Leonardo, Ideogram, Luma Ray). Utilized n8n automation workflows, API development with Postman, and MCP (Model Context Protocol) to build end-to-end autonomous solutions. Demonstrated expertise in prompt engineering techniques and LLM evaluation metrics to ensure high-quality agents

➢ AI RESEARCH INTERN at National Remote Sensing Centre, ISRO *May-July 2024*

- Developed a high-precision CNN model using TensorFlow/Keras, OpenCV, and NumPy, achieving 84.5% accuracy in classifying aquaculture ponds as "good" or "polluted." Integrated with a Random Forest pipeline to predict key water quality parameters. Utilized Google Earth Engine for satellite data extraction and processed geospatial data with rasterio, shapely, and pandas. Generated multiband raster images and interactive maps using Folium and Matplotlib for real-time pond health monitoring.

➢ Artificial Intelligence Intern with Hunar *April - 2024*

- Designed a rule-based chatbot using Python, NLTK, regular expressions, and finite state machines, and deployed it via Flask/Django for interactive communication. Additionally, built a weather prediction model using advanced regression techniques such as XGBoost, optimized through GridSearchCV for improved accuracy. Developed a spam detection system leveraging NLP with Naïve Bayes and SVM models, deployed as a RESTful API and integrated into a Streamlit app.

PROJECTS

➢ AI Add Video Generator | NagentAI *Feb 2025*

- Developed an AI-driven ad video generator that automates the creation of professional ad videos through a multi-stage process. The system begins with generating scene descriptions using OpenAI, followed by image creation from textual input (Ideogram). It then sets the scene with background elements, integrates voiceovers (ElevenLabs), adds background music, and finalizes the video (Runway). This project streamlined the production of engaging ad content, converting text into fully-realized videos.

➢ LinkedIn Post Generator with Image & Scheduling | NagentAI *Feb 2025*

- Developed a robust platform for automating LinkedIn post creation, complete with text generation and image integration. The platform leverages AI to generate relevant, engaging content and automatically pairs it with customized images. It also includes a scheduling feature, allowing users to plan and publish posts at optimal times, enhancing social media strategy and engagement. This all-in-one solution streamlines content management, making LinkedIn marketing more efficient and effective.

➢ SEO Content Optimizer | NagentAI *Feb 2025*

- Developed an SEO content optimizer that uses the Tavily Web Scraper API to extract website data for in-depth analysis. Users simply paste a website link to receive detailed insights into key SEO factors such as keyword density, meta descriptions, backlinks, page speed, and readability. The tool provides actionable recommendations for improving SEO, helping websites enhance their visibility and search engine rankings. By leveraging the Tavily API, this platform streamlines the process of content analysis.

➢ TweetPilot – Autonomous Twitter Engagement Agent

- Built a fully automated system using Google Gemini AI and Twitter API to monitor brand mentions, analyze sentiment, and post AI-generated replies in real time. Includes analytics dashboard (Next.js) and a scalable backend (NestJS, PostgreSQL, Redis, Docker).

➢ Customer Order Routing & Notification System | n8n *Mar 2025*

- Engineered a comprehensive workflow automation system using n8n that captures incoming customer bookings, instantly routes order details to appropriate departmental teams via email, generates automatic confirmations to customers, and delivers real-time status updates throughout the fulfillment process, significantly reducing response times while ensuring seamless communication.

➢ Intelligent HR Applicant Screening & Processing System | n8n *Mar 2025*

- Developed an efficient n8n-based recruitment automation system that collects applications from email and website submissions, filters candidates using customized parameters, and delivers qualified applicants to HR personnel, reducing screening time while maintaining consistent evaluation standards

Achievements and Certifications

➢ Certifications

- Advanced Certification in Data Science and Artificial Intelligence | CODE IIT MADRAS Pravartak, Intellipaat
- The Complete Data Structures and Algorithms in Python| Data Structures and algorithms |Python
- Python Programming Masterclass | Python | Udemy
- Bootcamp: Generative AI, LLM Apps, AI Agents, Cursor AI | AI Accelera

➢ Best Performance in Technology Award (2024-25) | IIT Kharagpur

- Honored for founding **CampusBuzz**, an innovative platform transforming student life by streamlining campus communication and engagement. Recognized for impactful use of technology, leadership, and contribution to the IIT Kharagpur community.
- Recognized as the academic topper of the department with a CGPA of 8.84.

➢ Positions Of Responsibility

- Led major student events like Diwali Illumination, Hall Day, and General Championship for MMM Hall, IIT Kharagpur (2100+ students), promoting leadership and campus engagement. Won a Gold Medal in the JNTU B-Zone Football Tournament. Also spearheaded placement efforts for the Coral Department, guiding students through resume building, interview prep, and industry.